

The Intersection of Antimicrobial-Resistant Tuberculosis and Social Vulnerabilities in Moldova: A Retrospective Analysis

Introduction:

Multidrug-resistant tuberculosis (MDR-TB), which was estimated to cause 13% of all antimicrobial resistance-attributable deaths globally, is defined as a strain of TB that is resistant to both rifampicin and isoniazid ⁽¹⁾. MDR-TB is more difficult to treat than regular TB and often results in worse patient outcomes, responsible for approximately 150,000 deaths annually since 2022 ^(2,3,4). Rifampicin and isoniazid are antibiotics used in both phases of the 6 to 9-month preferred TB treatment regimen, resulting in some MDR-TB patients undergoing alternative treatment that can last up to 24 months or longer ^(5,6).

Social factors (such as homelessness, incarceration status, and education) may also influence the risk of negative outcomes due to the lack of information, agency, and social protections in drug-resistant TB care ⁽⁷⁾. It is important to understand how strong the association between mortality and TB drug-resistance is, and any if relationship between social factors and TB drug-resistance exists, in order to best understand how to provide the highest level of care to MDR-TB patients. This project aims to quantify these relationships by testing two primary hypotheses: that all types of drug-resistant TB are associated with higher odds of mortality compared to drug-susceptible TB; and that social vulnerabilities are associated with higher odds of having some type of drug-resistant TB. By testing these hypotheses using binary and multinomial models, this project hopes to identify key socio-economic drivers and clinical outcomes of TB among this population.

Methods:

The data used in this project was obtained from a dataset provided by the Boston University School of Public Health. This dataset contains information on 11,687 TB positive patients from Moldova between 2008 to 2009. The information collected included, but was not limited to, demographics, infection characteristics, treatment, and outcome. To be included in this analysis patients had to have a confirmed TB test, drug susceptibility testing results, and a known treatment outcome. Patients were excluded from the analysis if they were missing data related to any of the inclusion criteria or predetermined social factors.

Association Between Drug Resistance & TB Mortality -

To evaluate the association between drug resistance and TB mortality a binary logistic regression was performed due to the primary outcome, TB mortality, being a binary variable. In this analysis, TB mortality was defined as death from TB progression during the study period. All other treatment outcomes were included in the comparison group. For this model the primary exposure of interest was TB drug resistance, a nominal variable. TB drug-resistance was categorized into 4 groups: drug-susceptible TB, mono-resistant TB, non-MDR poly-resistant TB, and MDR-TB. Drug-susceptible TB was used as the reference group. The covariates in this model were a set of 9 social factors; rural/urban, homeless, gender, age at diagnosis, citizenship, occupation, education, salaried, and ever incarcerated. These factors contain multiple variable types including continuous, binary, ordinal and nominal and were included to account for potential confounding in the association between drug resistance and TB mortality. These covariates were selected based on their potential to confound the relationship between drug resistant and mortality due to their association with treatment access and infection severity. Odds ratios and 95% confidence intervals were calculated to quantify the association.

Association Between Social Factors & Drug Resistance -

To evaluate the association between social factors and drug resistance multinomial logistic regression was performed, using drug-susceptible TB as the reference group. For this model, the primary outcome was drug resistance while the primary exposure of interest was the same social factors looked at in the previous model. Odds ratios and 95% confidence intervals were calculated to quantify the association between social factors and each drug resistance category relative to the reference group.

Results:

After applying inclusion and exclusion criteria to the initial dataset, the final study population consisted of 6,414 TB-positive patients in Moldova. The total population was predominantly male (75%) with a mean age at diagnosis of 39. When stratified by drug-resistance type, patients with MDR-TB had a higher proportion of high-risk social factors compared to all other drug-resistance types. Specifically, 70% of MDR-TB patients were

unemployed, 27% had been incarcerated previously, and 7.7% had experienced homelessness at some point in their life. The unadjusted mortality rate among MDR-TB patients was much higher than all other types at 12%, with the next closest group being mono-resistant TB patients at 6.5%.

Association Between Drug Resistance & TB Mortality –

The binary logistic regression model identified both drug-resistance type and social factors as significant predictors of mortality. After controlling for social factors, the model showed that MDR-TB patients had 3.4 times the odds of mortality (95% CI: 2.69 – 4.30) when compared to drug-susceptible TB patients. Mono-resistant TB patients had 1.76 times the odds of mortality (95% CI: 1.09 – 2.74) when compared to drug-susceptible TB patients.

While multiple social factors turned out to be significant predictors of mortality, it appears certain occupations are extremely strong predictors, specifically those who are unemployed (OR: 6.30 | 95% CI: 2.99 – 13.98), disabled (OR: 5.44 | 95% CI: 3.06 – 10.24), and pensioners (OR: 3.15 | 95% CI: 1.62 – 6.35) when compared to workers. People who have experienced homelessness were also at greater risk, with 2.01 times the odds of mortality (95% CI: 1.40 – 2.83) when compared to people who have never experienced homelessness. When looking at education, patients with a specialized secondary education had 0.59 times the odds of mortality (95% CI: 0.0037 – 0.89).

Association Between Social Factors & Drug Resistance –

The multinomial logistic regression model identified multiple social factors associated with the odds of having some type of drug-resistant TB when compared to drug-susceptible TB, however, only gender had consistent significance across all drug-resistance types. Being a female was associated with 26% lower odds of having MDR-TB (95% CI: 0.63 – 0.87), 40% lower odds of having mono-resistant TB (95% CI: 0.45 – 0.81), and 45% lower odds of having poly-resistant TB (95% CI: 0.39 – 0.77) relative to males. Interestingly, patients with no education also showed significantly reduced odds of contracting MDR-TB (OR: 0.50 | 95% CI: 0.30 – 0.84) when compared to patients with a secondary education.

Patients who have previously been incarcerated had 2.16 times the odds of contracting MDR-TB (95% CI: 1.85 – 2.54), the drug-resistance type most associated with mortality, when compared to patients who have never been incarcerated. Other social factors that significantly increase the odds of contracting MDR-TB compared to drug-susceptible TB include being on disability (OR: 1.93 | 95% CI: 1.49 – 2.51), having experienced homelessness (OR: 1.61 | 95% CI: 1.25 – 2.07), only having a primary education (OR 1.18 | 95% CI: 1.02 – 1.37), and living in an urban setting (OR: 1.15 | 95% CI: 1.02 – 1.31).

Patient Characteristics					
	Overall N = 6,414	MDR N = 1,464	Mono-Resistance N = 355	Poly-Resistance N = 268	Susceptible N = 4,327
Age at Diagnosis (Years)	39 [14]	38 [13]	40 [13]	40 [13]	40 [15]
Gender					
M	4,832 (75%)	1,187 (81%)	291 (82%)	224 (84%)	3,130 (72%)
F	1,582 (25%)	277 (19%)	64 (18%)	44 (16%)	1,197 (28%)
Type of Residence					
Rural	3,246 (51%)	684 (47%)	211 (59%)	144 (54%)	2,207 (51%)
Urban	3,168 (49%)	780 (53%)	144 (41%)	124 (46%)	2,120 (49%)
Salaried					
Y	1,977 (31%)	359 (25%)	101 (28%)	74 (28%)	1,443 (33%)
N	4,437 (69%)	1,105 (75%)	254 (72%)	194 (72%)	2,884 (67%)
Citizenship					
Moldova	6,382 (100%)	1,458 (100%)	353 (99%)	267 (100%)	4,304 (99%)
Other	32 (0.5%)	6 (0.4%)	2 (0.6%)	1 (0.4%)	23 (0.5%)
Ever Experienced Homelessness					
N	6,066 (95%)	1,351 (92%)	331 (93%)	254 (95%)	4,130 (95%)
Y	348 (5.4%)	113 (7.7%)	24 (6.8%)	14 (5.2%)	197 (4.6%)
Ever Incarcerated					
N	5,430 (85%)	1,076 (73%)	314 (88%)	230 (86%)	3,810 (88%)
Y	984 (15%)	388 (27%)	41 (12%)	38 (14%)	517 (12%)
Education Level					
Secondary School	3,718 (58%)	860 (59%)	215 (61%)	176 (66%)	2,467 (57%)
Primary School	1,445 (23%)	371 (25%)	83 (23%)	60 (22%)	931 (22%)
Specialized Secondary Education	913 (14%)	178 (12%)	41 (12%)	22 (8.2%)	672 (16%)
Higher Education	213 (3.3%)	35 (2.4%)	11 (3.1%)	8 (3.0%)	159 (3.7%)
No Education	125 (1.9%)	20 (1.4%)	5 (1.4%)	2 (0.7%)	98 (2.3%)
Occupation					
Worker	1,113 (17%)	184 (13%)	64 (18%)	37 (14%)	828 (19%)
Disabled	547 (8.5%)	156 (11%)	20 (5.6%)	25 (9.3%)	346 (8.0%)
Pensioner	419 (6.5%)	48 (3.3%)	17 (4.8%)	14 (5.2%)	340 (7.9%)
Student	277 (4.3%)	47 (3.2%)	12 (3.4%)	8 (3.0%)	210 (4.9%)
Unemployed	4,058 (63%)	1,029 (70%)	242 (68%)	184 (69%)	2,603 (60%)
TB Mortality					
All Other Treatment Outcomes	6,037 (94%)	1,286 (88%)	332 (94%)	253 (94%)	4,166 (96%)
Death From TB Progression	377 (5.9%)	178 (12%)	23 (6.5%)	15 (5.6%)	161 (3.7%)

* mean [SD]; n (%)

Table 1.
Baseline Characteristics of Tuberculosis Patients by Drug Resistance Status

Association Between Drug Resistance & TB Mortality		
	OR	(95% CI)
Drug Resistance		
Susceptible	1.00	Reference
MDR	3.40	(2.69 – 4.30)
Mono-Resistance	1.76	(1.09 – 2.74)
Poly-Resistance	1.41	(0.78 – 2.37)
Demographics		
Age at Diagnosis (1 year change)	1.03	(1.02 – 1.04)
Male	1.00	Reference
Female	1.11	(0.84-1.46)
Rural	1.00	Reference
Urban	1.06	(0.85 – 1.32)
Salaried	1.00	Reference
Not Salaried	0.65	(0.36 – 1.12)
Moldavian Citizenship	1.00	Reference
Other Citizenship	1.00	(0.16 – 3.54)
Never Homeless	1.00	Reference
Has Experienced Homelessness	2.01	(1.40 – 2.83)
Never Incarcerated	1.00	Reference
Previously Incarcerated	1.27	(0.96 – 1.68)
Education Level		
Secondary School	1.00	Reference
Primary School	1.38	(1.08 – 1.75)
Specialized Secondary Education	0.59	(0.037 – 0.89)
Higher Education	0.44	(0.10 – 1.21)
No Education	0.89	(0.31 – 2.08)
Occupation		
Worker	1.00	Reference
Student	1.24	(0.19 – 4.89)
Pensioner	3.15	(1.62 – 6.35)
Disabled	5.44	(3.06 – 10.24)
Unemployed	6.30	(2.99 – 13.98)

Table 2.
Adjusted Odds Ratios for the Association Between Drug Resistance and Tuberculosis Mortality

Association Between Drug Resistance & Social Factors						
(Susceptible as Reference)	MDR		Mono-Resistance		Poly-Resistance	
	OR	(95% CI)	OR	(95% CI)	OR	(95% CI)
Demographics						
Age at Diagnosis (1 year change)	0.99	(0.98 - 1.00)	1.00	(0.99 - 1.01)	1.00	(0.99 - 1.01)
Rural	1.00	Reference	1.00	Reference	1.00	Reference
Urban	1.15	(1.02 - 1.31)	0.73	(0.58 - 0.92)	0.97	(0.75 - 1.25)
Never Homeless	1.00	Reference	1.00	Reference	1.00	Reference
Has Experienced Homelessness	1.61	(1.25 - 2.07)	1.59	(1.01 - 2.49)	1.17	(0.66 - 2.06)
Male	1.00	Reference	1.00	Reference	1.00	Reference
Female	0.74	(0.63 - 0.87)	0.60	(0.45 - 0.81)	0.55	(0.39 - 0.77)
Moldavian Citizenship	1.00	Reference	1.00	Reference	1.00	Reference
Other Citizenship	0.61	(0.24 - 1.54)	0.98	(0.23 - 4.22)	0.67	(0.09 - 5.06)
Salaried	1.00	Reference	1.00	Reference	1.00	Reference
Not Salaried	1.20	(0.86 - 1.68)	0.69	(0.36 - 1.34)	0.77	(0.38 - 1.59)
Never Incarcerated	1.00	Reference	1.00	Reference	1.00	Reference
Previously Incarcerated	2.16	(1.85 - 2.54)	0.83	(0.58 - 1.17)	0.98	(0.68 - 1.41)
Education						
Secondary Education	1.00	Reference	1.00	Reference	1.00	Reference
Primary School	1.18	(1.02 - 1.37)	1.03	(0.79 - 1.35)	0.92	(0.67 - 1.25)
Specialized Secondary Education	0.85	(0.70 - 1.03)	0.76	(0.53 - 1.07)	0.51	(0.32 - 0.80)
Higher Education	0.91	(0.61 - 1.34)	1.00	(0.52 - 1.92)	0.98	(0.46 - 2.07)
No Education	0.50	(0.30 - 0.84)	0.64	(0.25 - 1.63)	0.29	(0.07 - 1.22)
Occupation						
Worker	1.00	Reference	1.00	Reference	1.00	Reference
Student	0.69	(0.43 - 1.09)	1.08	(0.47 - 2.48)	1.10	(0.42 - 2.94)
Pensioner	0.80	(0.55 - 1.18)	0.65	(0.35 - 1.21)	1.02	(0.50 - 2.09)
Unemployed	1.17	(0.81 - 1.71)	1.57	(0.78 - 3.19)	1.83	(0.83 - 4.03)
Disabled	1.93	(1.49 - 2.51)	0.70	(0.41 - 1.20)	1.52	(0.89 - 2.60)

Table 3.
Adjusted Odds Ratios for the Association Between Drug Resistance and Social Factors

Discussion:

As hypothesized, the biological and clinical complexity of drug-resistant TB strains translates directly to worse patient outcomes. Especially MDR-TB, whose patient's had more than triple the odds of mortality compared to patients with drug-susceptible TB, even after accounting for all social factors. This result really emphasizes the historical challenges associated with longer, more toxic, and less effective second-line treatments for TB. One of the most surprising findings from the mortality model is how impactful some of the social factors were. The greatest predictor, unemployment, increased the odds of mortality by 530% compared to employed workers. Being on disability was the second most impactful predictor, increasing the odds of mortality by more than 440% compared to employed workers. Both of these factors likely indicate that the individual faces profound structural barriers in life, making adherence to long and toxic drug regimens harder than it would be for the general population.

Certain social factors were also revealed to be key components in the contraction of drug-resistant TB. Having been incarcerated previously was the strongest social predictor for MDR-TB compared to drug-susceptible TB, more than doubling a patient's odds. This aligns with other literature that has identified prisons as high-transmission environments for TB due to overcrowding, poor ventilation, and poor access to healthcare. Being on disability was the second strongest predictor for MDR-TB. This can potentially be explained by the fact that disabled individuals are exposed to these TB strains more often, are increasingly susceptible to infections, or face significant barriers to accessing healthcare.

Two of this analysis' strengths come from the robustness of the dataset used (N = 6,414) and the low amount of missing data. This analysis, however, did not come without its limitations. Due to the retrospective nature of this study, it allows for the identification of associations and trends but does not allow for definitive causal claims. Additionally, the date from which this data was collected (2008 – 2009) was during the 2008 financial crisis which significantly impacted the global economy, including Moldova's economy. This may reduce the generalizability of certain social factors such as occupation and whether the patient is salaried or not.

Conclusion:

The intersection of antimicrobial-resistant TB and social vulnerabilities creates a compounding crisis that cannot be fixed with pharmacology alone. This analysis has demonstrated that while the presence of MDR-TB significantly increases the risk of mortality, the social realities of one's life may be just as, if not more, impactful. To effectively stop the spread of drug-resistant TB, public health practitioners must look beyond just clinical frameworks. It is vital for infection control and public health that social protection programs and economic support are available to the public.

Citations:

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